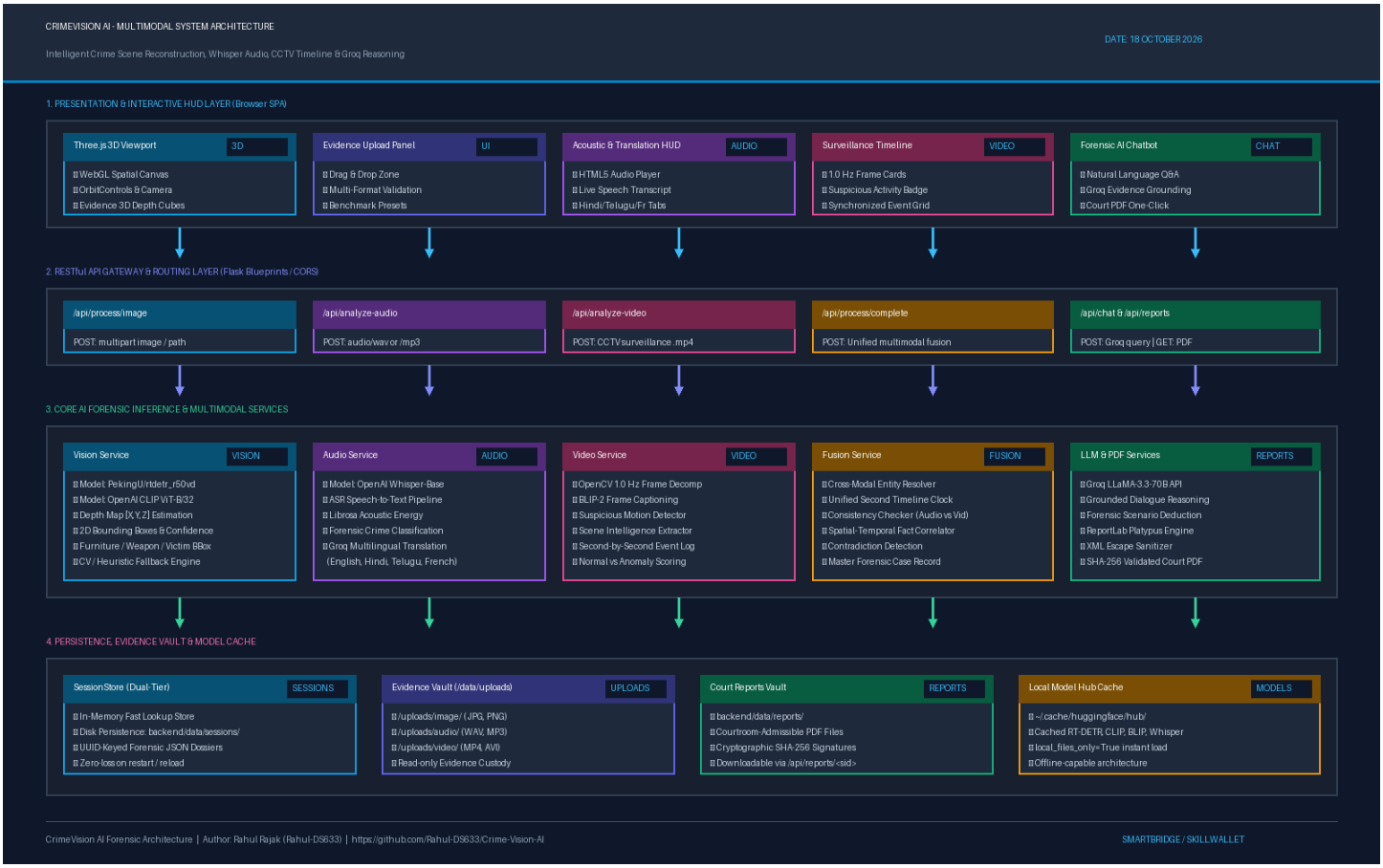
Solution Architecture

Date	18 October 2026
Team ID	Rahul-DS633
Project Name	CrimeVision AI - Intelligent Multimodal Crime Scene Reconstruction
Maximum Marks	5 Marks

1. System Architecture Diagram (High-Resolution Visual)



2. Architectural Layers & Component Breakdown

Layer	Components & Tech	Responsibilities & Communication
Presentation Layer (Client Browser)	• Three.js WebGL 3D Canvas • Vanilla ES6 Controllers • FontAwesome 6 / Dark CSS	Renders interactive 3D HUD, handles file drag-and-drop, orbit controls, and asynchronous API communication via fetch() in api.js.
API Gateway (Flask Blueprints)	• processing_bp (/api/process) • audio_analysis_bp • video_analysis_bp • chat_bp & results_bp	Validates incoming multipart/form-data and JSON payloads, enforces file size constraints, sanitizes filenames, and routes to backend services.
AI Services Layer (Core Engine)	• vision_service (RT-DETR, CLIP) • audio_service (Whisper ASR) • video_service (1Hz sampler) • fusion_service & pdf_service	Executes deep learning inference, extracts features, calculates 3D coordinate meshes, synthesizes multilingual text, and builds PDF reports.

Persistence Layer (Dual-Tier Store)	• SessionStore (In-memory dict) • Disk Cache (JSON files) • Reports Vault (/data/reports)	Maintains resilient state caching, historical session retrieval by session_id, and static serving of evidence files and generated court PDFs.